

FloodRisk FYRE — User Guide

A 3D flood-resilience game for a riverside campus. Make decisions, watch the flood rise, see what fails.

► [Play: final-eight-theta-85.vercel.app](https://final-eight-theta-85.vercel.app)

1. Open & basic controls

1. Open the link above in a desktop browser (Chrome, Safari, Edge, Firefox).
2. On the start screen, choose one of the three modes below.
3. In any 3D view: **drag** = orbit · **scroll / pinch** = zoom · **click a building** = scan it (damage, flood depth, power, loss).
4. To exit a mode, click **×** (top-left) or the **≈ logo** to return to the menu.

Reading the campus: lit windows = powered, dark windows = no power · cyan power lines = energized, red lines = failed · a red ⚡ **OUT / DARK** tag over a building means it lost power (DARK = dry but still lost power).

2. Live Drill ⌚ TIMED + SCORE

A timed emergency. The clock runs, the river rises, you act before deadlines.

1. **Prep:** choose storm severity (Minor → Extreme) and tick the capital upgrades to buy up front (drainage, elevate buildings, raise substation, elevate roads). Stay within budget.
2. Click **Start the drill**. The clock begins and water rises.
3. **Answer the alert cards** (deploy barriers, evacuate, sandbag the substation, close roads, open a shelter). Each has a shrinking timer — act before it runs out, or the option is lost. Acting **earlier** helps more.
4. Track the top bar: clock, water level, loss, people at risk, buildings dark, budget.
5. At the end you get a **resilience score (0–100) and grade**, broken down by loss avoided, people protected, lifelines, and budget.
6. Use **Re-run drill** to try other choices, or **Back to menu**.

Goal: highest score = lowest loss, most people protected, power kept on, within budget.

3. Decision Lab 🔄 COMPARE A vs B

Two tabs at top-right: **Guided dilemmas** and **Free compare**.

Guided dilemmas

1. Read the situation and the two plans (A and B).
2. Click a plan chip to view that plan's campus in 3D; click the other to flip.
3. **Predict:** click the plan you think is more resilient.
4. See both outcomes revealed side-by-side, plus the concept it teaches.
5. **Next dilemma** to continue, or **Retry**.

Free compare

1. Set **Plan A** (left) and **Plan B** (right): a flood scenario + any mitigations.
2. Click a panel header (**view in 3D**) to show that plan's campus.
3. Read the **scoreboard**: damage, buildings dark, people, downtime, spend, cost of risk. The better value in each row is highlighted, with a recommended plan and one-line insight.

4. Insurance Desk \$ PRICE RISK

Price the flood risk like an actuary and score the book. Data unlocks in **4 levels**: **1 Inventory** (flat rate only) → **2 Hazard** (PML) → **3 Vulnerability** (EAL → actuarially fair) → **4 Affordability** (caps → full score).

1. Click **Unlock Level N** → to reveal more data, then pick a **pricing strategy** (locked ones show the level they need).
2. Adjust the **levers**: loading factor, deductible, co-insurance.
3. Read the **Portfolio scorecard** (right): the 0–100 composite and its three parts — **Coverage** (premiums cover expected loss), **Affordability** (under the rate-on-line cap), **Profitability** (combined ratio < 100% = underwriting profit).
4. **Click any building** for its breakdown — EAL (pure premium), PML at 100/500-yr, premium, deductible, rate on line, loss & combined ratio. Green ring = premium covers its EAL; red = under-priced.

5. Sandbox 🎮 FREE PLAY

1. Pick a **flood scenario** (top-right).
2. Toggle **mitigations** (left) — the 3D campus and CAT dashboard update instantly.
3. Switch **role** (Emergency Manager, Engineer, Budget Officer, Risk Analyst) to change what the dashboard emphasizes.
4. **Click any building** to scan its details.
5. Click **Run before / after reflection** to compare your plan against doing nothing.

Risk = Hazard × Exposure × Vulnerability

You can't change the hazard (the flood), but you **can** change exposure and vulnerability. Watch for the **cascade**: if the power substation floods, every building it feeds goes dark — even dry ones on high ground. Often, protecting the lifeline (raising the substation, lowering the water) does more for resilience than reducing flood damage to individual buildings.

Educational tool with simplified, illustrative data — not a research-grade model.